Abhinav Gupta
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DBMS Lab 8

Task 1 - Database Setup

17)Create a MongoDB database named streamvault_db

```
test> use streamvault_db
switched to db streamvault_db
```

18)Create all four collections: users, movies, watch_history, ratings.

```
switched to db streamvault_db
streamvault_db> db.createCollection("users");
| db.createCollection("movies");
| db.createCollection("watch_history");
| db.createCollection("ratings");
{ ok: 1 }
streamvault_db>
```

19)Insert a minimum of 10 documents per collection. Ensure representation across multiple countries, genres, languages, subscription plans, and device types.

A)Users

CODE -

```
db.users.insertMany([
  { userId:"U1001", username:"Alice", email:"alice@example.com", country:"USA", subscriptionPlan:"Premium", subscriptionFee:15.99, joinedOn:ISODate("2023-01-15"), isActive:true, preferredGenres:["Thriller","Sci-Fi"], totalWatchTime:1200 },
  { userId:"U1002", username:"Raj", email:"raj@example.com", country:"India", subscriptionPlan:"Standard", subscriptionFee:9.99, joinedOn:ISODate("2022-11-10"), isActive:true, preferredGenres:["Drama","Comedy"], totalWatchTime:800 },
  { userId:"U1003", username:"Maria", email:"maria@example.com", country:"Brazil", subscriptionPlan:"Basic", subscriptionFee:5.99, joinedOn:ISODate("2021-07-20"), isActive:true, preferredGenres:["Romance","Action"], totalWatchTime:600 },
  { userId:"U1004", username:"Kenji", email:"kenji@example.com", country:"Japan", subscriptionPlan:"Premium", subscriptionFee:15.99, joinedOn:ISODate("2024-02-01"), isActive:true, preferredGenres:["Anime","Sci-Fi"], totalWatchTime:950 },
  { userId:"U1005", username:"Fatima", email:"fatima@example.com", country:"UAE", subscriptionPlan:"Standard", subscriptionFee:9.99, joinedOn:ISODate("2023-09-12"), isActive:true, preferredGenres:["Thriller","Drama"], totalWatchTime:400 },
  { userId:"U1006", username:"David", email:"david@example.com", country:"UK", subscriptionPlan:"Basic", subscriptionFee:5.99, joinedOn:ISODate("2020-05-05"), isActive:false, preferredGenres:["Comedy","Documentary"], totalWatchTime:300 },
  { userId:"U1007", username:"Li Wei", email:"liwei@example.com", country:"China", subscriptionPlan:"Premium", subscriptionFee:15.99, joinedOn:ISODate("2022-03-18"), isActive:true, preferredGenres:["Action","Sci-Fi"], totalWatchTime:1100 },
  { userId:"U1008", username:"Sofia", email:"sofia@example.com", country:"Spain", subscriptionPlan:"Standard", subscriptionFee:9.99, joinedOn:ISODate("2021-08-25"), isActive:true, preferredGenres:["Romance","Drama"], totalWatchTime:700 },
```

Abhinav Gupta
2401AI56
DBMS Lab 8

```
  { userId:"U1009", username:"Ahmed", email:"ahmed@example.com", country:"Egypt",
subscriptionPlan:"Basic", subscriptionFee:5.99, joinedOn:ISODate("2024-01-10"), isActive:true,
preferredGenres:["Action","Thriller"], totalWatchTime:500 },
  { userId:"U1010", username:"Elena", email:"elena@example.com", country:"Russia",
subscriptionPlan:"Premium", subscriptionFee:15.99, joinedOn:ISODate("2023-06-14"), isActive:true,
preferredGenres:["Sci-Fi","Fantasy"], totalWatchTime:900 }
]);

db.users.insertMany([
  { userId:"U1001", username:"Alina", email:"alina@example.com", country:"India",
subscriptionPlan:"Premium", subscriptionFee:15.99, joinedOn:ISODate("2024-01-15"), isActive:true,
preferredGenres:["Romance","Sci-Fi"], totalWatchTime:2200 },
  { userId:"U1002", username:"Raj", email:"raj@example.com", country:"India",
subscriptionPlan:"Premium", subscriptionFee:15.99, joinedOn:ISODate("2022-01-10"), isActive:true,
preferredGenres:["Drama","Comedy"], totalWatchTime:1800 },
  ]);
```

Output -

```
> db.users.insertMany([
    { userId:"U1001", username:"Alice", email:"alice@example.com", country:"USA", subscriptionPlan:"Premium", subscriptionFee:15.99, joinedOn:ISODate
    { userId:"U1002", username:"Raj", email:"raj@example.com", country:"India", subscriptionPlan:"Standard", subscriptionFee:9.99, joinedOn:ISODate("
    { userId:"U1003", username:"Maria", email:"maria@example.com", country:"Brazil", subscriptionPlan:"Basic", subscriptionFee:5.99, joinedOn:ISODate
    { userId:"U1004", username:"Kenji", email:"kenji@example.com", country:"Japan", subscriptionPlan:"Premium", subscriptionFee:15.99, joinedOn:ISODa
    { userId:"U1005", username:"Fatima", email:"fatima@example.com", country:"UAE", subscriptionPlan:"Standard", subscriptionFee:9.99, joinedOn:ISODa
    { userId:"U1006", username:"David", email:"david@example.com", country:"UK", subscriptionPlan:"Basic", subscriptionFee:5.99, joinedOn:ISODate("20
    { userId:"U1007", username:"Li Wei", email:"liwei@example.com", country:"China", subscriptionPlan:"Premium", subscriptionFee:15.99, joinedOn:ISOD
    { userId:"U1008", username:"Sofia", email:"sofia@example.com", country:"Spain", subscriptionPlan:"Standard", subscriptionFee:9.99, joinedOn:ISODa
    { userId:"U1009", username:"Ahmed", email:"ahmed@example.com", country:"Egypt", subscriptionPlan:"Basic", subscriptionFee:5.99, joinedOn:ISODate(
    { userId:"U1010", username:"Elena", email:"elena@example.com", country:"Russia", subscriptionPlan:"Premium", subscriptionFee:15.99, joinedOn:ISOD
  ]);
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('69c3b3b080adb5db1ec590df'),
      '1': ObjectId('69c3b3b080adb5db1ec590e0'),
      '2': ObjectId('69c3b3b080adb5db1ec590e1'),
      '3': ObjectId('69c3b3b080adb5db1ec590e2'),
      '4': ObjectId('69c3b3b080adb5db1ec590e3'),
      '5': ObjectId('69c3b3b080adb5db1ec590e4'),
      '6': ObjectId('69c3b3b080adb5db1ec590e5'),
      '7': ObjectId('69c3b3b080adb5db1ec590e6'),
      '8': ObjectId('69c3b3b080adb5db1ec590e7'),
      '9': ObjectId('69c3b3b080adb5db1ec590e8')
    }
  }
```

Abhinav Gupta
2401AI56
DBMS Lab 8

```
> db.users.insertMany([
    { userId:"U1001", username:"Alina", email:"alina@example.com", country:"India", subscriptionPlan:"Premium", subscriptionFee:15.99, joinedOn:ISODa
    { userId:"U1002", username:"Raj", email:"raj@example.com", country:"India", subscriptionPlan:"Premium", subscriptionFee:15.99, joinedOn:ISODate("
    ]);
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('69c3b7ca80adb5db1ec59107'),
      '1': ObjectId('69c3b7ca80adb5db1ec59108')
    }
  }
```

B)Movies

CODE -

db.movies.insertMany([
 { movieId:"M2001", title:"The Silent Sea", genre:["Sci-Fi","Thriller"], language:"Korean", releaseYear:2021, durationMinutes:120, director:"Choi Hang-yong", cast:["Bae Doona","Gong Yoo"], streamvaultRating:8.2, isOriginal:true, ageRating:"U/A" },
 { movieId:"M2002", title:"Lagaan", genre:["Drama","Sports"], language:"Hindi", releaseYear:2001, durationMinutes:224, director:"Ashutosh Gowariker", cast:["Aamir Khan","Gracy Singh"], streamvaultRating:9.0, isOriginal:false, ageRating:"U" },
 { movieId:"M2003", title:"RRR", genre:["Action","Drama"], language:"Telugu", releaseYear:2022, durationMinutes:182, director:"S.S. Rajamouli", cast:["Ram Charan","NTR Jr"], streamvaultRating:9.1, isOriginal:false, ageRating:"U/A" },
 { movieId:"M2004", title:"Gully Boy", genre:["Drama","Music"], language:"Hindi", releaseYear:2019, durationMinutes:153, director:"Zoya Akhtar", cast:["Ranveer Singh","Alia Bhatt"], streamvaultRating:8.5, isOriginal:false, ageRating:"U/A" },
 { movieId:"M2005", title:"Parasite", genre:["Thriller","Drama"], language:"Korean", releaseYear:2019, durationMinutes:132, director:"Bong Joon-ho", cast:["Song Kang-ho"], streamvaultRating:9.3, isOriginal:false, ageRating:"A" },
 { movieId:"M2006", title:"Dangal", genre:["Drama","Sports"], language:"Hindi", releaseYear:2016, durationMinutes:161, director:"Nitesh Tiwari", cast:["Aamir Khan"], streamvaultRating:8.8, isOriginal:false, ageRating:"U" },
 { movieId:"M2007", title:"Avengers: Endgame", genre:["Action","Sci-Fi"], language:"English", releaseYear:2019, durationMinutes:181, director:"Russo Brothers", cast:["Robert Downey Jr.","Chris Evans"], streamvaultRating:9.0, isOriginal:false, ageRating:"U/A" },
 { movieId:"M2008", title:"Article 15", genre:["Drama","Crime"], language:"Hindi", releaseYear:2019, durationMinutes:130, director:"Anubhav Sinha", cast:["Ayushmann Khurrana"], streamvaultRating:8.0, isOriginal:false, ageRating:"A" },
 { movieId:"M2009", title:"Interstellar", genre:["Sci-Fi"], language:"English", releaseYear:2014, durationMinutes:169, director:"Christopher Nolan", cast:["Matthew McConaughey"], streamvaultRating:9.2, isOriginal:false, ageRating:"U/A" },
 { movieId:"M2010", title:"Dil Bechara", genre:["Romance","Drama"], language:"Hindi", releaseYear:2020, durationMinutes:150, director:"Mukesh Chhabra", cast:["Sushant Singh Rajput"], streamvaultRating:7.5, isOriginal:false, ageRating:"U/A" }
]);

Output -

Abhinav Gupta
2401AI56
DBMS Lab 8

```
> db.movies.insertMany([
    { movieId:"M2001", title:"The Silent Sea", genre:["Sci-Fi","Thriller"], language:"Korean", releaseYear:2021, durationMinutes:120, director:"Choi
    { movieId:"M2002", title:"Lagaan", genre:["Drama","Sports"], language:"Hindi", releaseYear:2001, durationMinutes:224, director:"Ashutosh Gowarike
    { movieId:"M2003", title:"RRR", genre:["Action","Drama"], language:"Telugu", releaseYear:2022, durationMinutes:182, director:"S.S. Rajamouli", ca
    { movieId:"M2004", title:"Gully Boy", genre:["Drama","Music"], language:"Hindi", releaseYear:2019, durationMinutes:153, director:"Zoya Akhtar", c
    { movieId:"M2005", title:"Parasite", genre:["Thriller","Drama"], language:"Korean", releaseYear:2019, durationMinutes:132, director:"Bong Joon-ho
    { movieId:"M2006", title:"Dangal", genre:["Drama","Sports"], language:"Hindi", releaseYear:2016, durationMinutes:161, director:"Nitesh Tiwari", c
    { movieId:"M2007", title:"Avengers: Endgame", genre:["Action","Sci-Fi"], language:"English", releaseYear:2019, durationMinutes:181, director:"Rus
    { movieId:"M2008", title:"Article 15", genre:["Drama","Crime"], language:"Hindi", releaseYear:2019, durationMinutes:130, director:"Anubhav Sinha"
    { movieId:"M2009", title:"Interstellar", genre:["Sci-Fi"], language:"English", releaseYear:2014, durationMinutes:169, director:"Christopher Nolan
    { movieId:"M2010", title:"Dil Bechara", genre:["Romance","Drama"], language:"Hindi", releaseYear:2020, durationMinutes:150, director:"Mukesh Chha
  ]);
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('69c3b3fc80adb5db1ec590e9'),
      '1': ObjectId('69c3b3fc80adb5db1ec590ea'),
      '2': ObjectId('69c3b3fc80adb5db1ec590eb'),
      '3': ObjectId('69c3b3fc80adb5db1ec590ec'),
      '4': ObjectId('69c3b3fc80adb5db1ec590ed'),
      '5': ObjectId('69c3b3fc80adb5db1ec590ee'),
      '6': ObjectId('69c3b3fc80adb5db1ec590ef'),
      '7': ObjectId('69c3b3fc80adb5db1ec590f0'),
      '8': ObjectId('69c3b3fc80adb5db1ec590f1'),
      '9': ObjectId('69c3b3fc80adb5db1ec590f2')
    }
  }
```

C)watch_history

CODE -

```
db.watch_history.insertMany([
  { userId:"U1001", movieId:"M2001", watchedOn:ISODate("2024-05-10T20:00:00Z"),
watchDurationMinutes:110, completionPercent:92.0, deviceType:"Laptop", country:"USA" },
  { userId:"U1002", movieId:"M2002", watchedOn:ISODate("2024-06-12T18:30:00Z"),
watchDurationMinutes:200, completionPercent:89.0, deviceType:"Mobile", country:"India" },
  { userId:"U1003", movieId:"M2004", watchedOn:ISODate("2024-07-01T19:00:00Z"),
watchDurationMinutes:150, completionPercent:95.0, deviceType:"TV", country:"Brazil" },
  { userId:"U1004", movieId:"M2007", watchedOn:ISODate("2024-07-15T21:00:00Z"),
watchDurationMinutes:170, completionPercent:85.0, deviceType:"TV", country:"Japan" },
  { userId:"U1005", movieId:"M2008", watchedOn:ISODate("2024-08-05T16:00:00Z"),
watchDurationMinutes:40, completionPercent:25.0, deviceType:"Mobile", country:"UAE" },
  { userId:"U1006", movieId:"M2009", watchedOn:ISODate("2024-09-10T14:00:00Z"),
watchDurationMinutes:60, completionPercent:35.0, deviceType:"Laptop", country:"UK" },
  { userId:"U1007", movieId:"M2003", watchedOn:ISODate("2024-10-01T20:00:00Z"),
watchDurationMinutes:180, completionPercent:98.0, deviceType:"TV", country:"China" },
  { userId:"U1008", movieId:"M2006", watchedOn:ISODate("2024-09-10T14:00:00Z"),
watchDurationMinutes:161, completionPercent:35.0, deviceType:"Laptop", country:"India" },
  { userId:"U1009", movieId:"M20010", watchedOn:ISODate("2024-10-01T20:00:00Z"),
watchDurationMinutes:150, completionPercent:98.0, deviceType:"Laptop", country:"India" },
  { userId:"U1010", movieId:"M2005", watchedOn:ISODate("2024-10-01T20:00:00Z"),
watchDurationMinutes:132, completionPercent:98.0, deviceType:"TV", country:"Korea" }
  ]);
```

Abhinav Gupta
2401AI56
DBMS Lab 8

Output -

```
> db.watch_history.insertMany([
    { userId:"U1001", movieId:"M2001", watchedOn:ISODate("2024-05-10T20:00:00Z"), watchDurationMinutes:110, completionPercent:92.0, deviceType:"Lapto
    { userId:"U1002", movieId:"M2002", watchedOn:ISODate("2024-06-12T18:30:00Z"), watchDurationMinutes:200, completionPercent:89.0, deviceType:"Mobil
    { userId:"U1003", movieId:"M2004", watchedOn:ISODate("2024-07-01T19:00:00Z"), watchDurationMinutes:150, completionPercent:95.0, deviceType:"TV",
    { userId:"U1004", movieId:"M2007", watchedOn:ISODate("2024-07-15T21:00:00Z"), watchDurationMinutes:170, completionPercent:85.0, deviceType:"TV",
    { userId:"U1005", movieId:"M2008", watchedOn:ISODate("2024-08-05T16:00:00Z"), watchDurationMinutes:40, completionPercent:25.0, deviceType:"Mobile
    { userId:"U1006", movieId:"M2009", watchedOn:ISODate("2024-09-10T14:00:00Z"), watchDurationMinutes:60, completionPercent:35.0, deviceType:"Laptop
    { userId:"U1007", movieId:"M2003", watchedOn:ISODate("2024-10-01T20:00:00Z"), watchDurationMinutes:180, completionPercent:98.0, deviceType:"TV",
    { userId:"U1008", movieId:"M2006", watchedOn:ISODate("2024-09-10T14:00:00Z"), watchDurationMinutes:161, completionPercent:35.0, deviceType:"Lapto
    { userId:"U1009", movieId:"M20010", watchedOn:ISODate("2024-10-01T20:00:00Z"), watchDurationMinutes:150, completionPercent:98.0, deviceType:"Lapt
    { userId:"U1010", movieId:"M2005", watchedOn:ISODate("2024-10-01T20:00:00Z"), watchDurationMinutes:132, completionPercent:98.0, deviceType:"TV",
    ]);

< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('69c3b52f80adb5db1ec590f3'),
      '1': ObjectId('69c3b52f80adb5db1ec590f4'),
      '2': ObjectId('69c3b52f80adb5db1ec590f5'),
      '3': ObjectId('69c3b52f80adb5db1ec590f6'),
      '4': ObjectId('69c3b52f80adb5db1ec590f7'),
      '5': ObjectId('69c3b52f80adb5db1ec590f8'),
      '6': ObjectId('69c3b52f80adb5db1ec590f9'),
      '7': ObjectId('69c3b52f80adb5db1ec590fa'),
      '8': ObjectId('69c3b52f80adb5db1ec590fb'),
      '9': ObjectId('69c3b52f80adb5db1ec590fc')
    }
  }
```

D)ratings

CODE

```
db.ratings.insertMany([
  { userId:"U1001", movieId:"M2001", rating:8.5, review:"Loved the suspense!",
ratedOn:ISODate("2024-05-11"), helpful:12 },
  { userId:"U1002", movieId:"M2002", rating:9.2, review:"Epic storytelling and music.",
ratedOn:ISODate("2024-06-13"), helpful:20 },
  { userId:"U1003", movieId:"M2004", rating:8.8, review:"Powerful performances.",
ratedOn:ISODate("2024-07-02"), helpful:15 },
  { userId:"U1004", movieId:"M2004", rating:9.0, review:"Best Hindi film in years.",
ratedOn:ISODate("2024-07-16"), helpful:18 },
  { userId:"U1005", movieId:"M2007", rating:9.5, review:"Marvel at its peak!",
ratedOn:ISODate("2024-08-06"), helpful:30 },
  { userId:"U1006", movieId:"M2009", rating:8.0, review:"Mind-bending but slow at times.",
ratedOn:ISODate("2024-09-11"), helpful:10 },
  { userId:"U1007", movieId:"M2003", rating:9.3, review:"RRR is a masterpiece!",
ratedOn:ISODate("2024-10-02"), helpful:25 },
  { userId:"U1008", movieId:"M2003", rating:9.0, review:"Action sequences blew my mind.",
ratedOn:ISODate("2024-10-03"), helpful:22 },
  { userId:"U1009", movieId:"M2005", rating:9.4, review:"Parasite is genius.",
ratedOn:ISODate("2024-11-01"), helpful:40 },
  { userId:"U1010", movieId:"M2010", rating:7.8, review:"Emotional but predictable.",
ratedOn:ISODate("2024-12-15"), helpful:8 }
]);
```

Abhinav Gupta
2401AI56
DBMS Lab 8

Output -

```
> db.ratings.insertMany([
    { userId:"U1001", movieId:"M2001", rating:8.5, review:"Loved the suspense!", ratedOn:ISODate("2024-05-11"), helpful:12 },
    { userId:"U1002", movieId:"M2002", rating:9.2, review:"Epic storytelling and music.", ratedOn:ISODate("2024-06-13"), helpful:20 },
    { userId:"U1003", movieId:"M2004", rating:8.8, review:"Powerful performances.", ratedOn:ISODate("2024-07-02"), helpful:15 },
    { userId:"U1004", movieId:"M2004", rating:9.0, review:"Best Hindi film in years.", ratedOn:ISODate("2024-07-16"), helpful:18 },
    { userId:"U1005", movieId:"M2007", rating:9.5, review:"Marvel at its peak!", ratedOn:ISODate("2024-08-06"), helpful:30 },
    { userId:"U1006", movieId:"M2009", rating:8.0, review:"Mind-bending but slow at times.", ratedOn:ISODate("2024-09-11"), helpful:10 },
    { userId:"U1007", movieId:"M2003", rating:9.3, review:"RRR is a masterpiece!", ratedOn:ISODate("2024-10-02"), helpful:25 },
    { userId:"U1008", movieId:"M2003", rating:9.0, review:"Action sequences blew my mind.", ratedOn:ISODate("2024-10-03"), helpful:22 },
    { userId:"U1009", movieId:"M2005", rating:9.4, review:"Parasite is genius.", ratedOn:ISODate("2024-11-01"), helpful:40 },
    { userId:"U1010", movieId:"M2010", rating:7.8, review:"Emotional but predictable.", ratedOn:ISODate("2024-12-15"), helpful:8 }
  ]);
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('69c3b54580adb5db1ec590fd'),
      '1': ObjectId('69c3b54580adb5db1ec590fe'),
      '2': ObjectId('69c3b54580adb5db1ec590ff'),
      '3': ObjectId('69c3b54580adb5db1ec59100'),
      '4': ObjectId('69c3b54580adb5db1ec59101'),
      '5': ObjectId('69c3b54580adb5db1ec59102'),
      '6': ObjectId('69c3b54580adb5db1ec59103'),
      '7': ObjectId('69c3b54580adb5db1ec59104'),
      '8': ObjectId('69c3b54580adb5db1ec59105'),
      '9': ObjectId('69c3b54580adb5db1ec59106')
    }
  }
```

Task 2 Query Implementation

20. Display all StreamVault Original movies rated above 7.5, sorted by streamvaultRating descending. Show only title, genre, releaseYear, and streamvaultRating.

CODE -

```
db.movies.find(
  { isOriginal: true, streamvaultRating: { $gt: 7.5 } },{ _id: 0, title: 1, genre: 1, releaseYear: 1, streamvaultRating: 1 }).sort({ streamvaultRating: -1 });
```

Output -

```
> db.movies.find(
    { isOriginal: true, streamvaultRating: { $gt: 7.5 } },{ _id: 0, title: 1, genre: 1, releaseYear: 1, streamvaultRating: 1 }).sort({ streamvaultRat
< {
    title: 'The Silent Sea',
    genre: [
      'Sci-Fi',
      'Thriller'
    ],
    releaseYear: 2021,
    streamvaultRating: 8.2
```

21. Display all movies that belong to both 'Thriller' and 'Drama' genres (both must be present). Use the $all operator.

CODE -

Abhinav Gupta
2401AI56
DBMS Lab 8

```
db.movies.find(
  { genre: { $all: ["Thriller", "Drama"] } },
  { _id: 0, title: 1, genre: 1, releaseYear: 1, streamvaultRating: 1 }
);
```

Output -

```
> db.movies.find(
    { genre: { $all: ["Thriller", "Drama"] } },
    { _id: 0, title: 1, genre: 1, releaseYear: 1, streamvaultRating: 1 }
  );
< {
    title: 'Parasite',
    genre: [
      'Thriller',
      'Drama'
    ],
    releaseYear: 2019,
    streamvaultRating: 9.3
  }
```

Abhinav Gupta
2401AI56
DBMS Lab 8

22)Display users from India with a Premium subscription plan who are active, sorted by totalWatchTime descending.

CODE-

```
db.users.find(
  {country: "India",subscriptionPlan: "Premium",isActive: true},
  { _id: 0,userId: 1,username: 1,email: 1, totalWatchTime: 1 }
).sort({ totalWatchTime: -1 });
```

Output -

```
> db.users.find(
    {country: "India",subscriptionPlan: "Premium",isActive: true},
    { _id: 0,userId: 1,username: 1,email: 1, totalWatchTime: 1 }
  ).sort({ totalWatchTime: -1 });
< {
    userId: 'U1001',
    username: 'Alina',
    email: 'alina@example.com',
    totalWatchTime: 2200
  }
  {
    userId: 'U1002',
    username: 'Raj',
    email: 'raj@example.com',
    totalWatchTime: 1800
  }
```

Abhinav Gupta
2401AI56
DBMS Lab 8

23. Display all watch_history records where completionPercent is greater than 80% and deviceType is 'TV'.

CODE-

```
db.watch_history.find({completionPercent :{$gt : 80},  deviceType :"TV"},
{_id:0,userId: 1, movieId: 1, completionPercent: 1, deviceType: 1, watchedOn: 1 })
```

Output -

```
> db.watch_history.find({completionPercent :{$gt : 80},  deviceType :"TV"},
  {_id:0,userId: 1, movieId: 1, completionPercent: 1, deviceType: 1, watchedOn: 1 })
< {
    userId: 'U1003',
    movieId: 'M2004',
    watchedOn: 2024-07-01T19:00:00.000Z,
    completionPercent: 95,
    deviceType: 'TV'
  }
  {
    userId: 'U1004',
    movieId: 'M2007',
    watchedOn: 2024-07-15T21:00:00.000Z,
    completionPercent: 85,
    deviceType: 'TV'
  }
  {
    userId: 'U1007',
    movieId: 'M2003',
    watchedOn: 2024-10-01T20:00:00.000Z,
    completionPercent: 98,
    deviceType: 'TV'
  }
  {
    userId: 'U1010',
    movieId: 'M2005',
    watchedOn: 2024-10-01T20:00:00.000Z,
    completionPercent: 98,
    deviceType: 'TV'
```

24)Find all movies released after 2018 that have a duration greater than 150 minutes and are available in the 'Hindi' language.

CODE -

```
db.movies.find({releaseYear: { $gt: 2018 },durationMinutes: { $gt: 150 },language: "Hindi"},
{ _id: 0, title: 1, genre: 1, releaseYear: 1, durationMinutes: 1, language: 1 });
```

Abhinav Gupta
2401AI56
DBMS Lab 8

Output -

```
> db.movies.find({releaseYear: { $gt: 2018 },durationMinutes: { $gt: 150 },language: "Hindi"},
  { _id: 0, title: 1, genre: 1, releaseYear: 1, durationMinutes: 1, language: 1 });
< {
    title: 'Gully Boy',
    genre: [
      'Drama',
      'Music'
    ],
    language: 'Hindi',
    releaseYear: 2019,
    durationMinutes: 153
  }
```

25)Determine the most popular genre on the platform by counting how many times movies of each genre appear in watch_history. Use $lookup to join watch_history with movies, then $unwind the genre array.

CODE-

```
db.watch_history.aggregate([{$lookup: {
from: "movies",localField: "movieId",foreignField: "movieId",as: "movieDetails"
}},
  {$unwind: "$movieDetails" },{$unwind: "$movieDetails.genre" },
  {$group:{_id: "$movieDetails.genre",watchCount: { $sum: 1 }}},
  {$sort:{watchCount: -1}},{$limit : 1}]);
```

Output -

```
> db.watch_history.aggregate([{$lookup: {
  from: "movies",localField: "movieId",foreignField: "movieId",as: "movieDetails"
  }},
    {$unwind: "$movieDetails" },{$unwind: "$movieDetails.genre" },
    {$group:{_id: "$movieDetails.genre",watchCount: { $sum: 1 }}},
    {$sort:{watchCount: -1}},{$limit : 1}]);
< {
    _id: 'Drama',
    watchCount: 6
  }
```

26)Calculate the average user rating for each movie using the ratings collection. Display movieId and avgRating, sorted by avgRating descending. Show only movies with at least 2 ratings.

CODE-

```
db.ratings.aggregate([
  {$group: {_id: "$movieId",avgRating:{$avg:"$rating"},ratingCount:{$sum:1}}},
{$match:{ratingCount:{ $gte: 2 } } },
```

```
{$sort:{avgRating:-1}},
{ $project:{_id: 0,movieId: "$_id",avgRating: 1}
}]);
```

Output -

```
> db.ratings.aggregate([
    {$group: {_id: "$movieId",avgRating:{$avg:"$rating"},ratingCount:{$sum:1}}},
  {$match:{ratingCount:{ $gte: 2 } } },
    {$sort:{avgRating:-1}},
    { $project:{_id: 0,movieId: "$_id",avgRating: 1}
    }]);
< {
    avgRating: 9.15,
    movieId: 'M2003'
  }
  {
    avgRating: 8.9,
    movieId: 'M2004'
  }
```

27)Identify the top 5 most active users based on total watch time. Display userId, username (via $lookup with users), and totalWatchTime.

CODE -

```
db.users.aggregate([
  {$sort: {totalWatchTime: -1 }},{$limit: 5},
  {$project:{_id: 0,userId: 1,username: 1,totalWatchTime: 1}}]);
```

Output-

```
{
  totalWatchTime: 200,
  userId: 'U1002',
  username: 'Raj'
}
{
  totalWatchTime: 200,
  userId: 'U1002',
  username: 'Raj'
}
{
  totalWatchTime: 180,
  userId: 'U1007',
  username: 'Li Wei'
}
{
  totalWatchTime: 170,
  userId: 'U1004',
  username: 'Kenji'
}
{
  totalWatchTime: 161,
  userId: 'U1008',
  username: 'Sofia'
}
```

28)Display the total number of watch events and total watch duration grouped by country of stream, for the year 2024.

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DBMS Lab 8

CODE-

```
db.watch_history.aggregate([
  {$match: {watchedOn:{$gte:ISODate("2024-01-01T00:00:00Z"),$lt:ISODate("2025-01-01T00:00:00Z")}}},
  {$group: {_id: "$country",totalEvents: { $sum: 1 },totalDuration: { $sum: "$watchDurationMinutes" }}},
  {$project:{_id: 0,country: "$_id",totalEvents: 1,totalDuration: 1}},
  {$sort:{totalEvents: -1}}
]);
```

Output -

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DBMS Lab 8

```
{
  totalEvents: 3,
  totalDuration: 511,
  country: 'India'
}
{
  totalEvents: 1,
  totalDuration: 132,
  country: 'Korea'
}
{
  totalEvents: 1,
  totalDuration: 60,
  country: 'UK'
}
{
  totalEvents: 1,
  totalDuration: 110,
  country: 'USA'
}
{
  totalEvents: 1,
  totalDuration: 150,
  country: 'Brazil'
}
{
  totalEvents: 1,
  totalDuration: 40,
  country: 'UAE'
```

```
{
  totalEvents: 1,
  totalDuration: 180,
  country: 'China'
}
{
  totalEvents: 1,
  totalDuration: 170,
  country: 'Japan'
}
```

29)Identify movies where the average completion percent across all users is below 40% (indicating potential drop-off or poor engagement).

CODE -

```
db.watch_history.aggregate([
  {$group:{_id:"$movieId", avgCompletion:{$avg:"$completionPercent"}}},
  {$match:{avgCompletion:{$lt:40 }}},
  {$lookup:{from:"movies",localField: "_id",foreignField: "movieId",as: "movieDetails"}},
{$unwind:"$movieDetails" },{$project:{_id: 0,movieId:"$_id",title: "$movieDetails.title",avgCompletion:1}},
  {$sort:{avgCompletion:1}}]);
```

Output -

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2401AI56
DBMS Lab 8

```
> db.watch_history.aggregate([
    {$group:{_id:"$movieId", avgCompletion:{$avg:"$completionPercent"}}},
    {$match:{avgCompletion:{$lt:40 }}},
    {$lookup:{from:"movies",localField: "_id",foreignField: "movieId",as: "movieDetails"}},
  {$unwind:"$movieDetails" },{$project:{_id: 0,movieId:"$_id",title: "$movieDetails.title",avgCompletion:1}},
    {$sort:{avgCompletion:1}}]);
< {
    avgCompletion: 25,
    movieId: 'M2008',
    title: 'Article 15'
  }
  {
    avgCompletion: 35,
    movieId: 'M2006',
    title: 'Dangal'
  }
  {
    avgCompletion: 35,
    movieId: 'M2009',
    title: 'Interstellar'
  }
```

30)Using $lookup, join ratings with movies to find the top-rated movie in each genre. For each genre, display the movie title and its average rating.

CODE -

```
db.ratings.aggregate([
  {$lookup:{from: "movies",localField: "movieId",foreignField: "movieId",as: "movieDetails"
  }},
  { $unwind: "$movieDetails" },{$unwind:"$movieDetails.genre"},
   {$group:{_id: { genre: "$movieDetails.genre", movieId: "$movieId", title: "$movieDetails.title"
},avgRating: { $avg: "$rating" }}},
  {$sort:{"_id.genre": 1, avgRating: -1 }},
  {$group:{_id:"$_id.genre",topMovie:{$first: "$_id.title"},avgRating:{$first:"$avgRating" }}},
  {$project:{_id:0,genre: "$_id",topMovie:1,avgRating:1}}]);
```

Output -

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2401AI56
DBMS Lab 8

```
< {
    topMovie: 'Avengers: Endgame',
    avgRating: 9.5,
    genre: 'Sci-Fi'
  }
  {
    topMovie: 'Gully Boy',
    avgRating: 8.9,
    genre: 'Music'
  }
  {
    topMovie: 'Avengers: Endgame',
    avgRating: 9.5,
    genre: 'Action'
  }
  {
    topMovie: 'Parasite',
    avgRating: 9.4,
    genre: 'Drama'
  }
  {
    topMovie: 'Lagaan',
    avgRating: 9.2,
    genre: 'Sports'
  }
  {
    topMovie: 'Parasite',
    avgRating: 9.4,
    genre: 'Thriller'
```

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2401AI56
DBMS Lab 8

31)Compute the total monthly subscription revenue for StreamVault in 2024. Assume each active user contributes their subscriptionFee per month. Use $group with $month and $year.

CODE-

```
db.users.aggregate([
  { $match: { isActive: true } },
  { $group: { _id: null, monthlyRevenue: { $sum: "$subscriptionFee" } } },
  { $project: { _id: 0, monthlyRevenue: 1, year: 2024 } }
]);
```

Output -

```
> db.users.aggregate([
    { $match: { isActive: true } },
    { $group: { _id: null, monthlyRevenue: { $sum: "$subscriptionFee" } } },
    { $project: { _id: 0, monthlyRevenue: 1, year: 2024 } }
  ]);
< {
    monthlyRevenue: 137.89000000000001
  }
```

32. Perform a genre-based recommendation query: For a given user (userId: 'U1002'), find movies in their preferredGenres that they have NOT yet watched. Cross-reference watch_history and movies collections.

CODE-

```
db.users.aggregate([
  {$match: { userId: "U1002" } },
  {$lookup: {from: "movies",localField: "preferredGenres",foreignField: "genre",as:
"recommendedMovies"}},
  {$unwind: "$recommendedMovies" },
  {$lookup: {from: "watch_history",let: { uid: "$userId", mid: "$recommendedMovies.movieId" },pipeline:
[{$match: { $expr: { $and: [
{$eq:["$userId", "$$uid"]},{$eq:["$movieId", "$$mid"]}]}}}],as: "watched"}},{$match:{watched:{$size:
0}}},{$group: {_id: "$recommendedMovies.movieId",
     title: { $first: "$recommendedMovies.title" },genre: { $first: "$recommendedMovies.genre"
},releaseYear: { $first: "$recommendedMovies.releaseYear"}}},
  {$project:{_id: 0,movieId: "$_id",title: 1,genre: 1,releaseYear:1}}]);
```

Output -

Output -

```
{
  userId: 'U1002',
  recommendedTitle: 'RRR',
  genre: [
    'Action',
    'Drama'
  ],
  releaseYear: 2022
}
{
  userId: 'U1002',
  recommendedTitle: 'Gully Boy',
  genre: [
    'Drama',
    'Music'
  ],
  releaseYear: 2019
}
{
  userId: 'U1002',
  recommendedTitle: 'Parasite',
  genre: [
    'Thriller',
    'Drama'
  ],
  releaseYear: 2019
}
{
  userId: 'U1002',
```

```
  userId: 'U1002',
  recommendedTitle: 'Dangal',
  genre: [
    'Drama',
    'Sports'
  ],
  releaseYear: 2016
}
{
  userId: 'U1002',
  recommendedTitle: 'Article 15',
  genre: [
    'Drama',
    'Crime'
  ],
  releaseYear: 2019
}
{
  userId: 'U1002',
  recommendedTitle: 'Dil Bechara',
  genre: [
    'Romance',
    'Drama'
  ],
  releaseYear: 2020
}
```